

Working with the LT-LiDER ProMut Platform

In a web browser, open the ProMut Platform <https://promut.uab.cat> and create an account/log in (top right, you will need a Google account).

Download the ZIP folder that came with this exercise and unzip it. It contains files we will use throughout this activity.

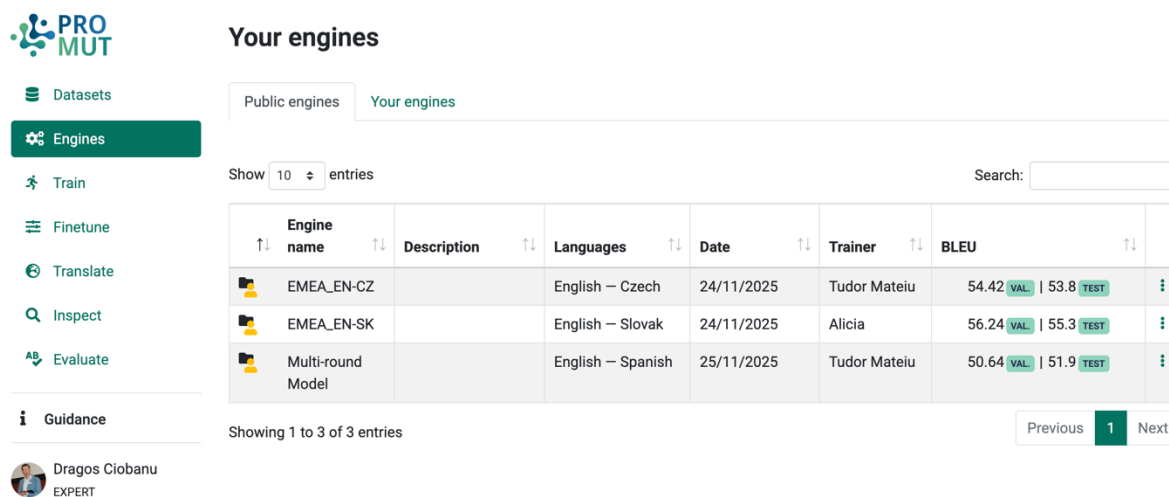
We will explore 3 scenarios:

1. Translate, inspect, and evaluate using existing MT engines (beginner user)
2. (DEMO – Expert user) Train a machine translation (MT) engine using a small open-source downloaded corpus;
 - a. (further explanation) Train an MT engine using personal TMX files
3. Fine-tune an existing engine with a downloaded dataset

1. 1. Translate

In order to translate, you need an MT engine. You can either *grab* (i.e. borrow) one that is already present in ProMut, or you can *train* a new engine. You will be grabbing one in this first exercise.

- a. Click on the **Engines** menu item on the left side of the ProMut interface
- b. Select the **Public engines** tab. The engines available will be displayed below as in the following screenshot:



PRO MUT

Datasets

Engines

Train

Finetune

Translate

Inspect

Evaluate

Guidance

Dragos Ciobanu
EXPERT

Your engines

Public engines | Your engines

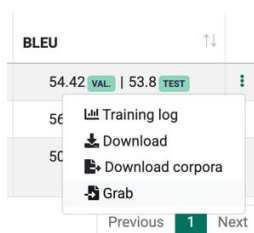
Show 10 entries Search:

Engine name	Description	Languages	Date	Trainer	BLEU
EMEA_EN-CZ		English – Czech	24/11/2025	Tudor Mateiu	54.42 VAL 53.8 TEST
EMEA_EN-SK		English – Slovak	24/11/2025	Alicia	56.24 VAL 55.3 TEST
Multi-round Model		English – Spanish	25/11/2025	Tudor Mateiu	50.64 VAL 51.9 TEST

Showing 1 to 3 of 3 entries

Previous 1 Next

- c. Click on the three vertical dots at the end of the line with your preferred MT engine and, from the drop-down menu, select **Grab**



BLEU

54.42 VAL | 53.8 TEST

56.24 VAL | 55.3 TEST

50.64 VAL | 51.9 TEST

Training log

Download

Download corpora

Grab

Previous 1 Next

- d. Now your chosen engine will appear in the **Your engines** tab

Your engines

[Public engines](#)
[Your engines](#)

Show 10 entries

Search:

Engine name	Description	Languages	Date	Trainer	BLEU
EMEA_EN-SK		English – Slovak	24/11/2025	Alicia	56.24 VAL 55.3 TEST

Showing 1 to 1 of 1 entries

Previous 1 Next

- Now click on the **Translate** menu option on the left
- Your engine will be part of the drop-down list at the top. If you have chosen several engines, they will all be listed in the drop-down menu
- On your computer desktop, from the downloaded workshop files, open a source file in English from your chosen language pair and copy the first 5 lines
- Paste these English source lines into ProMut (in the left-hand side box), then click on the Translate button below it
- The translation will appear in the right-hand box. It can also be downloaded as a TMX file.
- You can also translate entire files. Please translate the entire source English file for your language pair and save it on your computer.

1. 2. Inspect

- Click on the **Inspect** menu item on the left-hand side
- Make sure you are in the **Details** tab
- Copy-paste the first sentence of the English source file for your language pair
- Make sure your grabbed engine is selected, then click on **Inspect**
- ProMut will now display the source and target sentences being tokenised, as well as the n-best list

1. 3. Evaluate

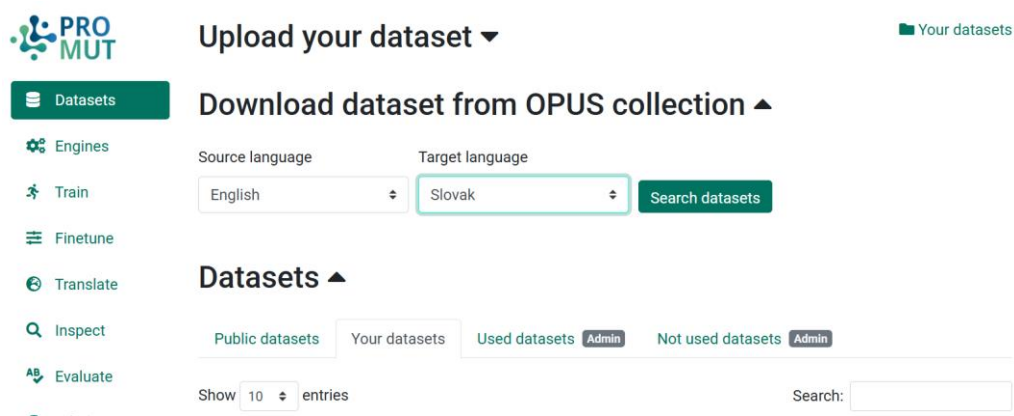
- For evaluation, you will need:
 - the source file in English,
 - the output in your chosen target language which you saved in step 1.1
 - a reference translation which you will find in the **reference_translations/** workshop file folder
- In ProMut/**Evaluate**, upload the files above to the relevant fields
- Click on **Evaluate**
- You will now see the automatic metrics which ProMut computes, and if you scroll to the bottom of the page, you will be able to see the scores alongside the source, target, and reference segments one by one.

2. (DEMO) Training a machine translation (MT) engine

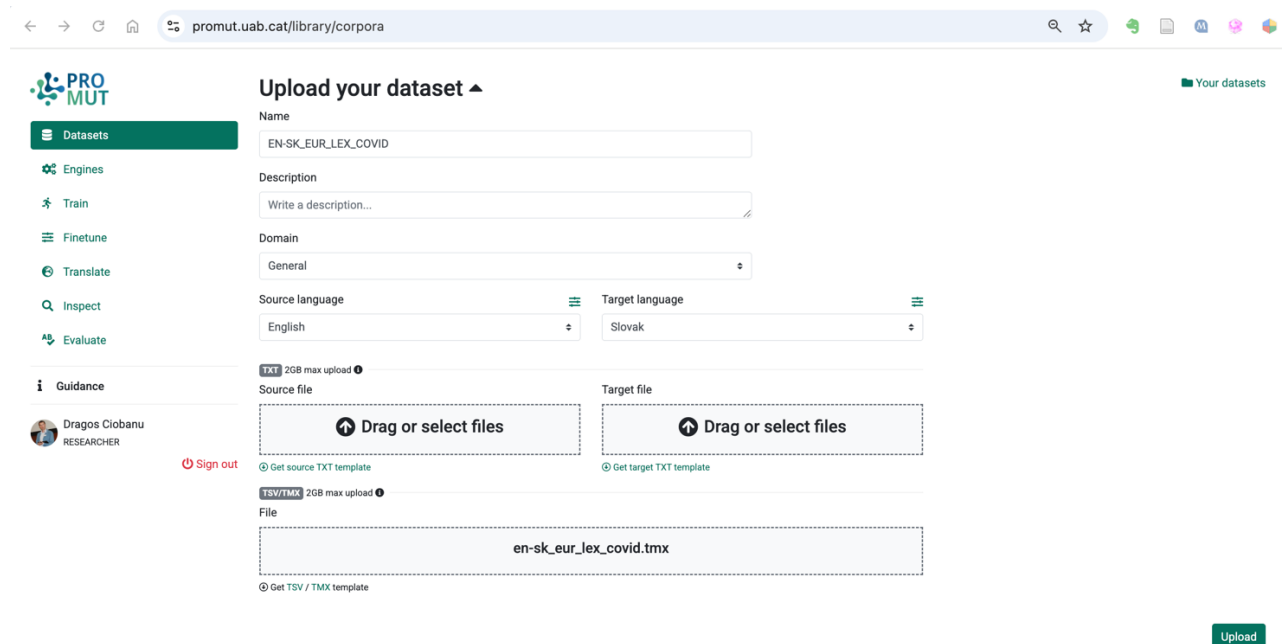
In order to train an engine, you need a bilingual corpus. You can download readily available corpora, one such repository being OPUS.

In the ProMut interface:

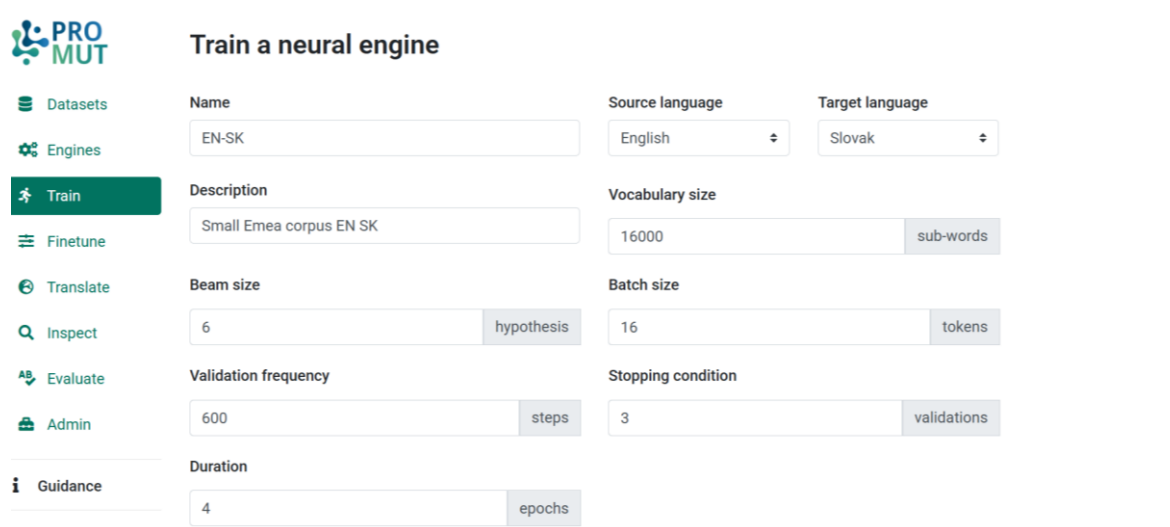
1. To download a corpus, go to the **Datasets** menu item
2. Expand the drop-down area for **Download Dataset from Opus Collection** by clicking on the arrow to the right of the title
 - a. Select source and target languages, and click on **Search datasets**
3. In the new table,
 - a. Select your chosen dataset, then click **Download**
 - b. The downloaded dataset will appear in the **Your datasets** tab



4. Alternatively, you can train an engine using a dataset which you upload. To upload, you will need to expand the **Upload your dataset** area in the screenshot above, give your dataset a name, select source and target languages, and upload either one TMX file or two parallel aligned source and target files in TXT format.



5. To train an engine using the OPUS downloaded dataset or your own uploaded dataset, select the **Train** menu item on the left
 - a. Give the engine a name
 - b. Select a source and target languages
 - c. You can also write a description for it
 - d. Keep the default values for the other parameters
 - i. When you have a big corpus, you can increase the default Duration from 1 epoch (in the Duration field)



PRO MUT

Train a neural engine

Left sidebar: Datasets, Engines, **Train**, Finetune, Translate, Inspect, Evaluate, Admin, Guidance

Form fields:

- Name:** EN-SK
- Source language:** English
- Target language:** Slovak
- Description:** Small Emea corpus EN SK
- Vocabulary size:** 16000 (sub-words)
- Beam size:** 6 (hypothesis)
- Batch size:** 16 (tokens)
- Validation frequency:** 600 (steps)
- Stopping condition:** 3 (validations)
- Duration:** 4 (epochs)

You are now ready to define the training data specification

- e. For the **Training set**, click in the box with the + sign next to it to write the number of segments to be used for the training from the total number available in the corpus (e.g., the corpus below eur_lex_covid_en-sk has a total of 22,478 from which we will chose 20,000 for training, 1,000 for Validation and 1,000 for Test).
- f. Click on the + sign to validate this.



ADMIN Sign out

Training set | **Validation set** | **Test set**

Search: []

Dataset	Count	Total
en-sk eur_lex_covid_en-sk	22478	22k total

Buttons: + (to validate)

- g. For the **Validation set**, write 1000 in the box next to + , then click on + to validate,
- h. Do the same for the **Test set**.
- i. Click on **Start training**

 Alina Secara
ADMIN

 Sign out

Training set
 eur_lex_covid_en-sk 
 20k 20k sentences - 89% of dataset
 2.5M

Validation set
 eur_lex_covid_en-sk 
 1k 1k sentences - 4% of dataset
 5k

Test set
 eur_lex_covid_en-sk 
 1k 1k sentences - 4% of dataset
 5k

Search:

Search:

Search:

6. A new window appears showing the training progress.



Datasets

Engines

Train

Finetune

Translate

Inspect

Evaluate

Training

EMEA_EN-SK

Engine settings

Vocabulary size	Beam size	Batch size
16000	6	16
Validation freq.	Stopping condition	Duration
600	3	4

Engine corpora

Training ELRC-EMEA (en-sk, 100k)	Validation ELRC-EMEA (en-sk, 1k)	Test ELRC-EMEA (en-sk, 1k)
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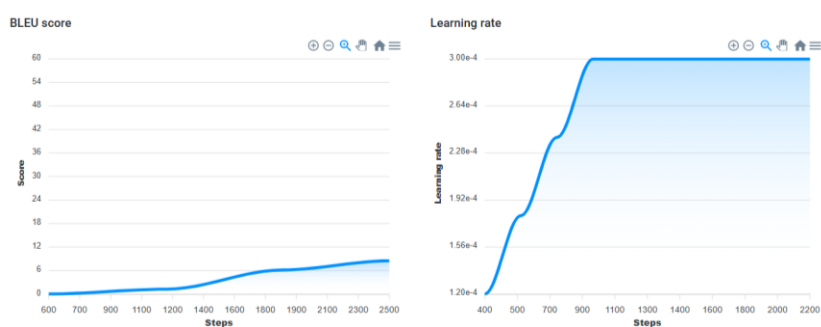
Stop

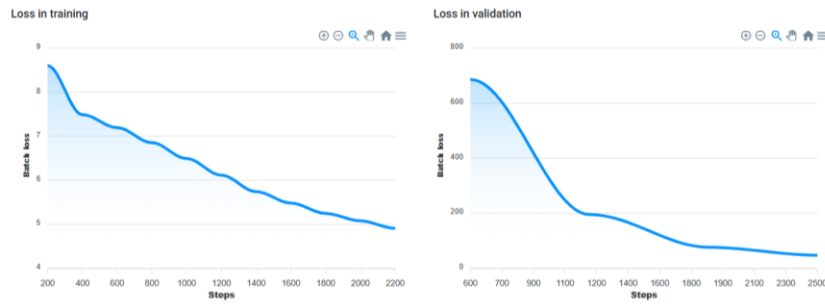
23:18:09

Epoch 1 out of 4e ⌵

0Wh ⚡

- During training, a clock shows the number of minutes, seconds (ignore the hours) and the epochs
- Further down, Loss in training - the lower the value, the higher the resemblance between the engine-generated sentences and the training corpus. As the training progresses, the line should go down;
- During training the BLEU score continuously computes the translation quality of the engine given the validation set. The aim is to select the engine with the highest computed BLEU score;
- The log of the learning process appears at the bottom of the page indicating how long the process took





7. Once the training is complete

- You get a summary of the training, including the time it took for the engine to be trained;
- In addition to the Validation BLEU score (which was used during the training), you can generate a BLEU for the Test corpus. Click on the Start test and a new BLEU will be generated.

Test zone

This engine scored 8.48444 BLEU points in validation.

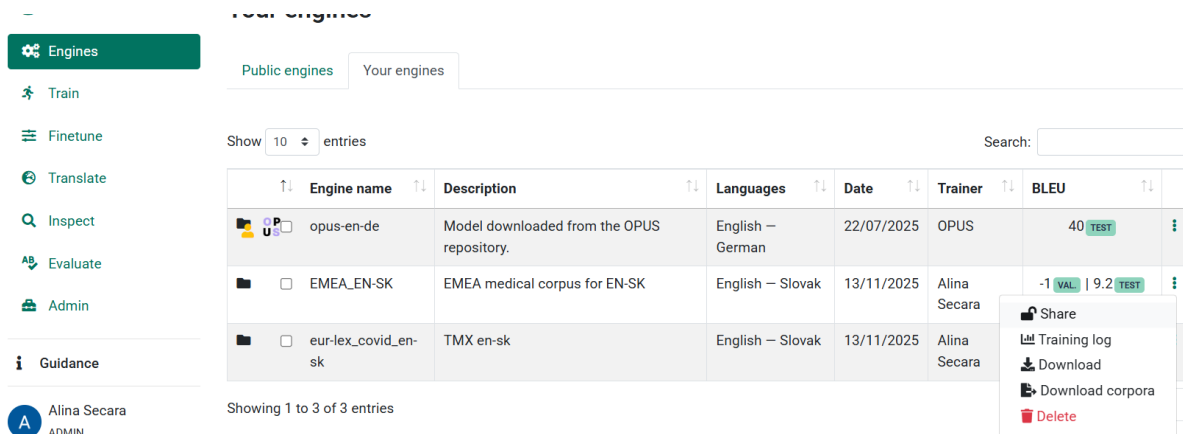
You can get a more accurate score by performing a translation test against the Test corpora.


9.20
BLEU

- When the engine finishes training (the epochs are finished) you can either Resume, to continue to train the engine, or use the engine to translate as you have already practised.

8. To use the engine, go to the **Engines** menu item

- Click on the **Your engines** tab
- Select your engine and click on the three vertical dots at the end of the line – you can Share it (it will be visible to other ProMut users, such as students).



Engine name	Description	Languages	Date	Trainer	BLEU
opus-en-de	Model downloaded from the OPUS repository.	English – German	22/07/2025	OPUS	40 TEST
EMEA_EN-SK	EMEA medical corpus for EN-SK	English – Slovak	13/11/2025	Alina Secara	-1 VAL 9.2 TEST
eur-lex_covid_en-sk	TMX en-sk	English – Slovak	13/11/2025	Alina Secara	

9. You can now use your engine to translate. Go to the **Translate** menu item on the left-hand side and either paste a source text or upload a document in your source language to be translated with your engine.

3. Fine-tune an existing engine with an OPUS downloaded dataset or your own translations in TMX or TXT format

When you do not have enough translations to train an engine from scratch, and these translations are in a specific domain, you could use them to fine-tune (adapt) a general engine to your specialised domain. In ProMut, you can fine-tune in two ways:

- By using a dataset from OPUS
- By uploading your own translations in TMX format or as two parallel text files

Fine-tuning with an OPUS dataset: first download the general OPUS engine, then fine-tune:

- Click on the **Engines** menu item
- Click on **Download models from OPUS collection**, select English and your target language, then click on **Search models**
- Download** the resulting engine displayed
- After it has been downloaded, the engine will appear in **Public engines** and you will need to **Grab** it so that it appears in the **Your engines** tab
- Click on the **Datasets** menu item
- Check the **Public datasets** tab under **Datasets**
 - Your chosen dataset for fine-tuning may have already been downloaded by another ProMut user
 - If it has not been downloaded already, download it from the **Download dataset from OPUS collection** area
 - If it has been downloaded, find it in the **Public datasets** tab and click on the three vertical dots next to its name to **Grab** it

English – Slovak						
ELRC-EMEA						
ELRC-EMEA dataset, downloaded from the OPUS collection						
File name		Language	Sentences	Words	Domain	Date
  ELRC-EMEA.en-sk.en	Preview Download Grab	English	780,097	13,467,345		24/11/2025
  ELRC-EMEA.en-sk.sk		Slovak	780,097	12,032,452		24/11/2025

- Click on the **Finetune** menu option on the left-hand side
- Give your engine an appropriate name – e.g. EMEA_EN-SK_FT
- Select the appropriate source and target languages, and your general OPUS engine which you have just downloaded and grabbed should appear in the drop-down list
- Keep the default parameters as they are, except the **Duration** – change that to 5 epochs
- In the **Training set** field, type 20000, then click on the + sign to validate your choice
- Use the same procedure to select 1000 segments for the **Validation set** and 1000 other segments for the **Test set**

promut.uab.cat/finetune/



- Datasets
- Engines
- Train
- Finetune**
- Translate
- Inspect
- Evaluate
- Guidance
- Dragos Ciobanu
RESEARCHER

Finetune a pre-trained neural engine

Name

Description

Vocabulary size
 sub-words

Batch size
 sentences

Stopping condition
 validations

Source language

Target language

Model

Beam size
 hypothesis

Validation frequency
 steps

Duration
 epochs

Sign out

Training set
ELRC-EMEA
20k 20k sentences - 3% of dataset

Validation set
ELRC-EMEA
1k 1k sentences - 0% of dataset

Test set
ELRC-EMEA
1k 1k sentences - 0% of dataset

Search:

Dataset

No datasets available for the selected languages

m. Click on **Start finetuning** and do not refresh the new web page – as fine-tuning progresses, this page will be updated with relevant information (have a look at the Training log at the end of the page, too).

promut.uab.cat/train/console/101



- Datasets
- Engines
- Train
- Finetune**
- Translate
- Inspect
- Evaluate
- Dragos Ciobanu
RESEARCHER

Training

EMEA_EN-SK_FT

Engine settings

Vocabulary size

Beam size

Batch size

Validation freq.

Stopping condition

Duration

Engine corpora

Training

Validation

Test

Engine stats

Time trained

Tokens per second

Score (Validation)

Sign out

00:01:57

Epoch 1 out of 5e

0Wh

Dashboard

Loss in training

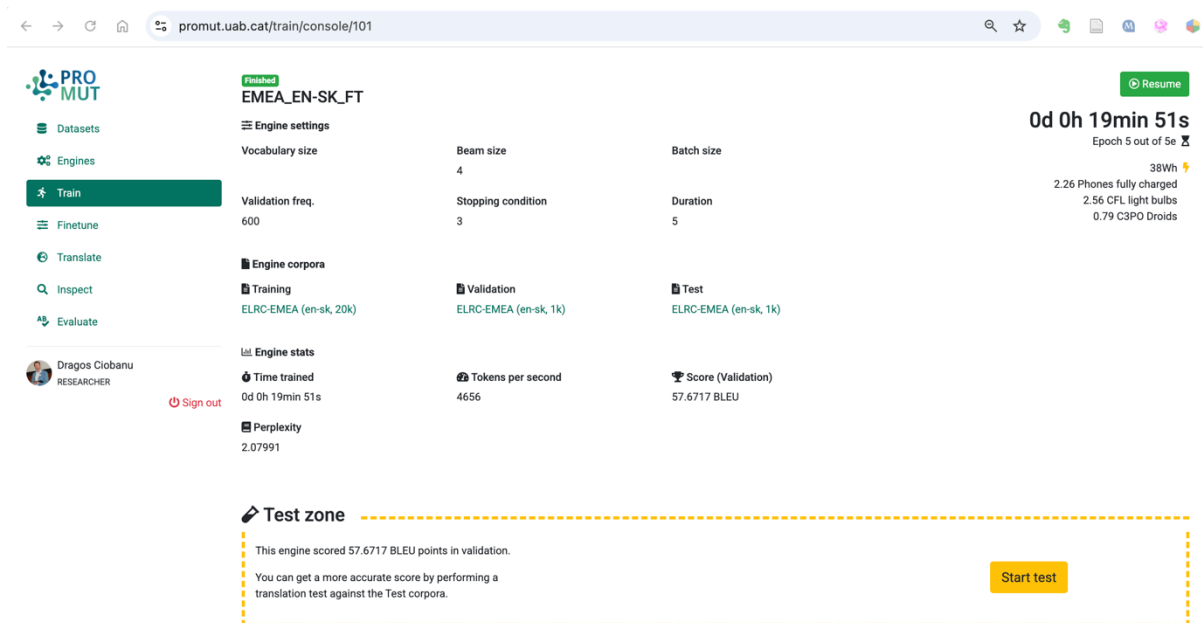
Loss in validation

n. After fine-tuning, ProMut will display the Validation score, and ask you to start the Test in the Test Zone to evaluate the performance of your newly fine-tuned engine against the test dataset



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Finished **EMEA_EN-SK_FT** Resume

0d 0h 19min 51s
Epoch 5 out of 5e

Engine settings

Vocabulary size	Beam size: 4	Batch size
Validation freq.: 600	Stopping condition: 3	Duration: 5

Engine corpora

Training ELRC-EMEA (en-sk, 20k)	Validation ELRC-EMEA (en-sk, 1k)	Test ELRC-EMEA (en-sk, 1k)
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Engine stats

Time trained 0d 0h 19min 51s	Tokens per second 4656	Score (Validation) 57.6717 BLEU
Perplexity 2.07991		

Test zone

This engine scored 57.6717 BLEU points in validation.
You can get a more accurate score by performing a translation test against the Test corpora. Start test

- o. After fine-tuning, you can use your new, specialised engine to translate, and you can also inspect and evaluate it as you have already learnt how to do.
- p. If you want to share this engine with your students so that they can test/inspect/evaluate it, you will need to find it in **Engines/Your engines/three-vertical-dots-near-your-engine/Share**